Python Libraries — Complete Notes

1. What Is a Library?

A **Library** is a collection of ready-made, reusable code that we can use in our own Python programs.

Instead of writing everything from the beginning, we can use code that other developers have already created.

Example

```python
import math

print(math.sqrt(25))
```

Output:

```text
5.0
```

Here:

```text
math      → Module
sqrt()    → Function
```

25 → Input

5.0 → Result

```

### Simple Definition

> A library is reusable code that helps us perform tasks without writing everything ourselves.

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## 2. Why Do We Use Libraries?

Libraries make programming easier and faster.

### 1. Code Reuse

We can use existing code instead of writing it again.

```python
import math

print(math.sqrt(100))
```

We don't need to create our own square-root function.

2. Faster Development
```

Libraries provide ready-made functions and tools.

For example:

```python
import random

print(random.randint(1, 10))
```

We can generate a random number without implementing the random-number algorithm ourselves.

3. Less Code

Without libraries, we may need to write many lines of code. With libraries, we can often perform the same task with just a few lines.

4. Tested Functionality

Popular libraries are used and tested by many developers. This can save development and testing time, although we should still understand and verify the library we use.

5. Easy Maintenance

Using well-designed libraries can make programs easier to maintain.

For example, instead of writing complex networking code ourselves, we can use a library such as `requests`.

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3. Module vs Package vs Library

These three terms are related, but they are not exactly the same.

What Is a Module?

A **Module** is usually a single Python file containing Python code.

Example:

```text
math_tools.py
```

It can contain:

```python
def add(a, b):
    return a + b
```

We can import it:

```python
import math_tools

print(math_tools.add(10, 20))
```

```

**Output:**

```text
30
```

Easy Definition

> Module = Usually one `.py` file.

---

4. What Is a Package?

A **Package** is a way to organize related Python modules together.

Example Structure

```text
myproject/
└── tools/
    ├── __init__.py
    ├── math_tools.py
    └── string_tools.py
```

Here:
```

```text
tools/          → Package
math_tools.py   → Module
string_tools.py → Module
```

Easy Definition

> Package = A collection or organization of related Python modules.

Note: Modern Python also supports namespace packages, so an `__init__.py` file is not always required.

---

5. What Is a Library?

A **Library** is a broader term for reusable code that developers can use in their programs.

A library may contain:

```text
Modules
Packages
Functions
Classes
Tools
```

Examples:

```text
NumPy
Pandas
Requests
Django
```

Easy Definition

> Library = Reusable code that provides useful functionality.

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6. Module vs Package vs Library

Term	Simple Meaning	Example
Module	Usually one Python file	`math_tools.py`
Package	Organizes related modules	`tools/`
Library	Reusable code for solving problems	NumPy, Pandas

Easy Memory Trick

```text
Module
    Usually one Python file
```

Package

    Organizes related modules

Library

    Provides reusable functionality
```

---

## 7. Python Standard Library

Python comes with a large collection of modules called the **Python Standard Library**.

You usually do not need to install these modules separately.

### Examples

```text
math
random
datetime
os
sys
json
re
```

7.1 math
```

The `math` module is used for mathematical operations.

```python
import math

print(math.sqrt(25))
print(math.pi)
```

Output:

```text
5.0
3.141592653589793
```

7.2 random

The `random` module is used to generate pseudorandom values.

```python
import random

print(random.randint(1, 10))
```

Possible output:

```text
7
```

The result can be different each time.

7.3 datetime

The `datetime` module is used to work with dates and times.

```python
from datetime import datetime

now = datetime.now()

print(now)
```

Possible output:

```text
2026-10-12 09:30:15.123456
```

The actual output depends on your system's date and time.

7.4 os

The `os` module is used for operating-system-related tasks.

```python
import os

print(os.getcwd())
```

This prints the current working directory.

7.5 sys

The `sys` module is used to interact with the Python interpreter and system-related information.

Example:

```python
import sys

print(sys.version)
```

It can also be used to access command-line arguments.

```python
import sys

print(sys.argv)
```

7.6 json

The `json` module is used to work with JSON data.

```python
import json

data = '{"name": "Faruk", "age": 25}'

result = json.loads(data)

print(result["name"])
```

Output:

```text
Faruk
```

Here, `json.loads()` converts a JSON-formatted string into a Python object.

7.7 re

The `re` module is used for **Regular Expressions**.

```python
import re
```

```
text = "My number is 12345"

result = re.findall(r"\d+", text)

print(result)
```

Output:

```text
['12345']
```

Here, `\d+` matches one or more digits.

---

8. Third-Party Libraries

Third-party libraries are developed outside Python's standard library. They often need to be installed separately.

Installation

```bash
pip install requests
```

Then import the library:

```python
import requests
```

---

9. Common Third-Party Python Libraries

Some popular Python libraries and frameworks are:

```text
NumPy
Pandas
Requests
Django
Flask
FastAPI
Scrapy
BeautifulSoup
Selenium
PyTorch
TensorFlow
```

These tools are used for different purposes, including web development, data analysis, web scraping, and machine learning.

NumPy

NumPy is mainly used for:

```text
Numerical computing
Arrays
Mathematical operations
Scientific computing
```

Example:

```python
import numpy as np

numbers = np.array([1, 2, 3, 4])

print(numbers)
```

Output:

```text
[1 2 3 4]
```

---

10. Pandas

Pandas is mainly used for:

```text
Data analysis
Data manipulation
Tables
CSV files
DataFrames
```

Example:

```python
import pandas as pd

data = {
    "Name": ["Alice", "Bob"],
    "Age": [20, 25]
}

df = pd.DataFrame(data)

print(df)
```

Output:

```text
    Name  Age
0  Alice   20
1    Bob   25
```

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11. Requests

Requests is a Python library used to send HTTP requests.

Example:

```python
import requests

response = requests.get("https://example.com")

print(response.status_code)
```

Output:

```text
200
```

The status code `200` generally means that the HTTP request succeeded.

The actual status code depends on the server's response.

Requests is commonly used for:

```text
APIs
Web requests
HTTP communication
Web data retrieval
```

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12. Django

Django is a Python web framework.

It is used to build:

```text
Web Applications
Websites
Backend Systems
REST APIs
```

Example concept:

```text
Django
   ↓
Python Web Framework
   ↓
Web Applications and Backend Systems
```

---

13. Flask

Flask is a lightweight Python web framework.

It can be used to create:

```text
Web Applications
REST APIs
Backend Services
```

Simple Example

```python
from flask import Flask

app = Flask(__name__)
```

```
@app.route("/")
def home():
    return "Hello World"
```

When the application is configured and running, visiting the `/` route returns `Hello World`.

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14. FastAPI

FastAPI is a modern Python framework mainly used for building APIs.

It is popular for:

```text
REST APIs
Backend Services
API Development
```

Example

```python
from fastapi import FastAPI

app = FastAPI()
```

```
@app.get("/")
def home():
    return {"message": "Hello World"}
```

When the application is running, a request to `/` returns a JSON response:

```json
{
    "message": "Hello World"
}
```

---

15. Scrapy

Scrapy is a Python framework used for web crawling and web scraping.

It can collect data from websites.

Common uses:

```text
Web Scraping
Web Crawling
Data Extraction
```

```

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## 16. BeautifulSoup

**BeautifulSoup** is a Python library used to parse HTML and XML documents.

It is commonly used to extract data from web pages.

### Example

```python
from bs4 import BeautifulSoup

html = "<h1>Hello</h1>"

soup = BeautifulSoup(html, "html.parser")

print(soup.h1.text)
```

Output:

```text
Hello
```

Here:
```

```text
BeautifulSoup     → Parses the HTML
html.parser       → HTML parser
soup.h1           → Finds the h1 element
soup.h1.text      → Extracts its text
```

---

17. Selenium

Selenium is used to automate web browsers.

It can perform actions such as:

```text
Open a browser
Click buttons
Fill forms
Navigate pages
Extract information
Test websites
```

It is commonly used for browser automation and website testing.

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18. PyTorch

PyTorch is a machine learning and deep learning framework.

It is commonly used for:

```text
Machine Learning
Deep Learning
Neural Networks
Computer Vision
Natural Language Processing
```

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19. TensorFlow

TensorFlow is another popular machine learning and deep learning framework.

It can be used for:

```text
Machine Learning
Deep Learning
Neural Networks
AI Applications
```

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20. Standard Library vs Third-Party Library

Standard Library	Third-Party Library
Comes with Python	Usually installed separately
Usually requires no separate installation	Often installed using `pip`
`math`	`numpy`
`random`	`pandas`
`datetime`	`requests`
`os`	`django`
`sys`	`flask`
`json`	`fastapi`
`re`	`scrapy`

Example: Standard Library

```python
import math

print(math.sqrt(25))
```

No separate installation is normally required.

Example: Third-Party Library

Install NumPy:

```bash
pip install numpy
```

Then import it:

```python
import numpy as np

numbers = np.array([1, 2, 3])

print(numbers)
```

---

21. How to Install a Third-Party Library

We commonly use `pip` to install Python packages.

General Syntax

```bash
pip install package_name
```

Example 1: Install Requests

```bash
pip install requests
```

Then:

```python
import requests
```

Example 2: Install NumPy

```bash
pip install numpy
```

Then:

```python
import numpy
```

Recommended Alternative

You can use the Python interpreter's `-m` option to run pip:

```bash
python -m pip install requests
```

This helps ensure that pip is associated with the Python interpreter you intend to use.

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22. Importing a Library

There are several common ways to import Python code.

22.1 Import the Whole Module

```python
import math

print(math.sqrt(25))
```

Here, we access the function through the module name.

22.2 Import with an Alias

```python
import numpy as np

numbers = np.array([1, 2, 3])

print(numbers)
```

Here:

```text
numpy    → Original module name
np       → Short alias
```

An alias makes the module name shorter to use.

22.3 Import a Specific Function

```python
from math import sqrt

print(sqrt(25))
```

We can call `sqrt()` directly without writing `math.sqrt()`.

22.4 Import Multiple Items

```python
from math import sqrt, pi

print(sqrt(25))
print(pi)
```

Output:

```text
5.0
3.141592653589793
```

---

23. Why Are Libraries Important in Python?

Python has a large ecosystem of libraries. Different libraries help us perform different tasks.

Web Development

```text
Django
Flask
FastAPI
```

Web Scraping and Browser Automation

```text
Scrapy
BeautifulSoup
Selenium
```

Data Science

```text
NumPy
Pandas
```

Machine Learning and Deep Learning

```text
PyTorch
TensorFlow
```

APIs and Backend Development

```text
Requests
Django REST Framework
FastAPI
```

Libraries allow developers to build applications without implementing every feature from scratch.

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24. Important Interview Questions

Q1. What is a Python library?

A Python library is reusable code that provides functions, classes, or tools for solving common programming problems.

Q2. Why do we use libraries?

The main reasons are:

```text
Code reuse
Faster development
Less code
Ready-made functionality
Easier maintenance
```

Q3. What is a module?

A module is usually a single Python file containing Python code.

Example:

```text
math_tools.py
```

Q4. What is a package?

A package is a way to organize related Python modules together.

Example:

```text
tools/
├── math_tools.py
└── string_tools.py
```

Q5. What is the Python Standard Library?

It is a collection of modules distributed with Python that can generally be used without installing separate third-party packages.

Examples:

```text
math
os
sys
json
re
datetime
```

Q6. What is a third-party library?

A third-party library is reusable Python code developed outside the Python Standard Library.

It is often installed using:

```bash
pip install package_name
```

Q7. What is pip?

`pip` is the standard package installer commonly used to install Python packages from the Python Package Index (PyPI).

Example:

```bash
python -m pip install requests
```

Q8. What is the difference between `import` and `from ... import`?

Using `import`:

```python
import math

print(math.sqrt(25))
```

We access the function through the module name.

Using `from ... import`:

```python
from math import sqrt

print(sqrt(25))
```

We import a specific function and access it directly.

Q9. What is the difference between a library and a framework?

A **library** provides reusable functionality that our code can call when needed.

A **framework** provides a structure for building an application and often controls the overall flow of execution.

Examples:

```text
Libraries:
NumPy, Pandas, Requests

Frameworks:
Django, Flask, FastAPI
```

The distinction is not always strict, and a tool may include both library-like and framework-like features.

Q10. Do we need to install every Python library?

No. Modules in the Python Standard Library generally come with Python. Third-party packages usually need to be installed separately.

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25. Quick Revision

Library

```text
Reusable code that provides useful functionality
```

Module

```text
Usually one .py file
```

Package

```text
Organizes related Python modules
```

Standard Library

```text
Comes with Python
```

Examples:

```text
math
os
sys
json
re
datetime
```

Third-Party Libraries

```text
Usually installed separately
```

Examples:

```text
NumPy
Pandas
Requests
BeautifulSoup
```

Selenium

PyTorch

TensorFlow

```

### Frameworks

```text

Django

Flask

FastAPI

Scrapy

```

Installation

```bash

pip install package_name

```

Import

```python

import package_name

```

---

26. Final Summary
```

The main idea is:

```text
Python Libraries
      ↓
Reusable Code
      ↓
Less Repeated Work
      ↓
Faster Development
      ↓
Easier Application Development
```

Easy Mental Model

```text
Module
   Usually one .py file

Package
   Organizes related modules

Library
   Provides reusable functionality

Framework
   Provides a structure for building applications
```

```

### Most Important Examples

**Standard Library:**

```python
import math

print(math.sqrt(25))
```

Output:

```text
5.0
```

Third-Party Library:

First install Requests:

```bash
pip install requests
```

Then:

```python
```
```

```
import requests

response = requests.get("https://example.com")

print(response.status_code)
```

One-Line Definition

> A Python library is reusable code that helps us perform tasks without writing everything from scratch.